

The AI-native, Secure, Interoperable Communication Infrastructure for Distributed Energy Resources

A Gateway enabling
Dispatch + Telemetry
for a smarter, self-reliant grid

• THE VISION

Every energy asset, visible and **dispatchable.**

A world where 100M+ distributed energy resources — inverters, batteries, EV chargers, meters — speak one common, secure language instead of a thousand proprietary ones.

4+ TW

of distributed energy devices, globally

"Every OEM built its own hive. Nobody built the meadow. We're building the meadow — the shared infrastructure every hive can safely connect to."

• THE MISSION

One gateway. Zero integration tax.

Ship a sub-\$50, pay-once edge gateway that connects, normalizes, dispatches, and verifies every DER — any OEM, open standards, agent-ready — so utilities, aggregators, and asset owners can harvest stranded flexibility instead of paying to unlock it.



Connect

On-site gateway speaks to every asset locally over open standards.
No vendor lock-in.



Normalize

Translates every device language into one clean API. No per-OEM work.



Dispatch

Charge, discharge, curtail, shift — from any platform, in real time.



Verify

Every dispatch event logged before/during/after. Provable, claimable.

● THE PROBLEM

Today, the hive is **fragmented**.

Every OEM locks its devices in a separate silo.

80%

of DER fleets still lack dispatchability, existing APIs are too expensive, and lack interoperability and security.

100M+ DERS DEPLOYED · NO COMMON DISPATCH LAYER · \$50K+ /MO INTEGRATION TAX PER OEM



● ● THE PROBLEM — SECURITY

One weak cell, and the **wasp** gets in.

A single compromised device shouldn't be able to threaten an entire fleet — but today, it can.

80% of DER fleets still lack dispatchability, existing APIs are too expensive, and lack interoperability and security.



● THE SOLUTION

Every asset guards itself.

Each DER sits in its own sealed cell. One compromised device never reaches the fleet.

- **Trusted Execution Environment** — policy enforced on the edge device
- **Zero-trust device identity** — each DER enrolled and authenticated alone
- **Data sovereignty** — telemetry stays in-jurisdiction, never offshore



● THE SOLUTION

One gateway speaks every grid language.

SOUTHBOUND — DEVICES

- IEEE 2030.5
- SunSpec Modbus
- MQTT
- Zigbee
- Custom integrations



OR

SOFTWARE LICENSE

IEEE 2030.5

NORTHBOUND — RECIPIENTS

- Utility
- Aggregators
- Integrators
- OEMs
- Financiers

● ARCHITECTURE

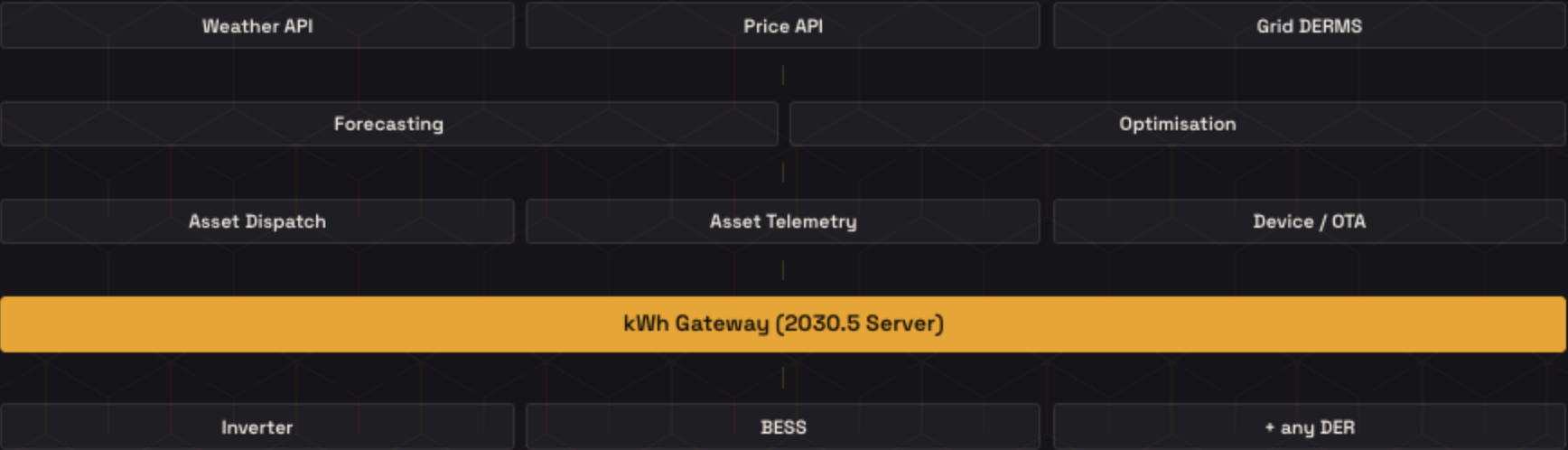
One path is closed.
One path is open.

TODAY — OEM-LOCKED PATH



One proprietary chokepoint per OEM.

WITH KWH — OPEN GATEWAY PATH



One open, secured gateway for every OEM.

● DER GATEWAY

Palm-sized. Under \$50 BOM.

PROTOCOL STACK

IEEE 2030.5 · SunSpec/Modbus · MQTT · Zigbee

DEVICES UNDER MANAGEMENT

Solar inverters · batteries · EV chargers · smart meters · flexible loads

BUSINESS MODEL

\$150 one-time gateway license + SaaS

Forecasting

P2P Trading

Demand Flex

Securitisation

Asset Health

INTEROPERABILITY + SECURITY + MCP

COMMUNICATION INFRASTRUCTURE

● APPLICATION STACK

Three layers, one **stack**.



SaaS App Store

Applications sold on top of the gateway, and gateway sold on top of applications.



Edge Gateway

Discovers, monitors, and dispatches DERs locally.



Protocol Translation

Speaks IEEE 2030.5 to utilities and aggregators.



Cloud Platform

Fleet management, provisioning, telemetry, analytics.

• WHO IT'S FOR

Six ways into the **hive**.

01

Utilities / DISCOMs

Can't see or dispatch behind-the-meter assets.
→ Full visibility + dispatch, no new integrations.

02

Aggregators / VPP Operators

Every asset type needs its own integration.
→ One API dispatches the whole fleet.

03

OEMs

Building a software platform is slow and expensive.
→ Instant interoperability, no cloud to build.

04

Energy AI Platforms

DER data is messy, proprietary, inconsistent.
→ Clean, agent-ready data + dispatch APIs.

05

BESS Financiers / Asset Owners

No verifiable telemetry to underwrite against.
→ Cryptographically verified telemetry.

06

Commercial Asset Owners

Assets sit idle, invisible to grid revenue.
→ Lower bills, new revenue, full visibility.

● USE CASE 01 — UTILITY / DISCOM

A DISCOM sees its rooftop fleet for the first time.

2,500+ distributed rooftop solar sites, 30+ MW — previously invisible behind the meter. The kWh gateway makes every site visible in real time and dispatchable for curtailment and flexibility during grid stress, without touching tariffs or settlement.

ZODIAC ENERGY PILOT · 2,500+ SITES · 30+ MW

BEFORE
No visibility. Manual estimates.

AFTER
Real-time telemetry + dispatch across the fleet.

● USE CASE 02 — AGGREGATOR / VPP

One virtual power plant, five different manufacturers.

A VPP operator managing batteries, inverters, EV chargers, and meters from different OEMs once needed a separate integration per vendor — at \$50K+/month tax each. With kWh, the whole mixed fleet dispatches through one gateway and one API.

KINTECH SYNERGY · LIVE BESS ASSETS · TELEMETRY + SOC + DISPATCH VALIDATION

BEFORE
N integrations, N months of engineering.

AFTER
One gateway, live in days.

● USE CASE 03 — FINANCIER

Provable telemetry, financeable assets.

BESS and DER financiers need independent, tamper-evident proof of asset performance to underwrite and securitize loans against distributed assets. kWh’s Verify step logs every dispatch event before, during, and after — turning raw telemetry into claimable, auditable revenue and lower-risk collateral.

ACTIVE ENGAGEMENT WITH FANNIE MAE ON VERIFIED TELEMETRY FOR MICROGRID/DATACENTER LOAN SECURITIZATION

BEFORE
Unverifiable, OEM-reported numbers.

AFTER
Cryptographically verified, in-jurisdiction telemetry trail.

• THE OPPORTUNITY

A \$30B+ market, inflecting now.

4 TWh of devices require security.



• WHERE WE'RE GOING

Global infrastructure. Local proof points.

kWh is global infrastructure — India is our beachhead channel for proving velocity, not our market cap. Primary targets: US, Canada, Australia, India. Secondary: UK, Germany, Netherlands.

SOURCED

United States

IEEE 2030.5 mandated under California Rule 21. FERC Order 2222 opens wholesale markets to aggregated DERs. Commercial center of gravity — HQ Palo Alto.

MARKET CONTEXT

Australia

One of the world's highest rooftop solar penetration rates per capita — creating acute grid-visibility and orchestration pressure on distribution networks.

SOURCED

India

2.6M rooftops / 9.6 GW added in ~2 years under PM Surya Ghar. Gujarat leads at 6.67 GW. Live channel into 5 DISCOMs via Beckn DEG — also reachable in Oman, Qatar, Brazil.

● UNIT ECONOMICS

The fix costs less than a day of the problem.



\$150 one-time gateway license + recurring SaaS. No per-dispatch fees. No lock-in.

One avoided month of integration tax $\approx$ 300+ gateways deployed (\$50,000 $\div$ \$150).

One foundation, many revenues — the same telemetry stream monetizes Demand Response, VPP, P2P Trading, Asset Health, Securitization.

● PILOT PARTNERS

A live channel
into **DISCOMs.**

Zodiac Energy

2,500+ sites, 30+ MW rooftop solar (Gujarat). Data sharing live; gateway telemetry pilot in setup.

Kintech Synergy

₹800Cr+ EPC, 30+ years solar/wind/storage (Gujarat). Live BESS assets for telemetry, state-of-charge, dispatch validation.

HARDWARE

PCB Power India — custom PCB for production

CHANNEL

Beckn DEG — 5 DISCOMs across India

DEPLOYING

Piloting — 2 plants, live since Dec 2025

● TIMELINE

From live pilots to global infrastructure.



Dec 2025

- MVP live — IEEE 2030.5 on production hardware
- 2 operating plants



Next 6 months (from Jul 2026)

- Ship \$50-BOM production gateway (custom PCB)
- Full protocol coverage — EV, batteries, inverters, appliances
- Fleet-scale discovery & enrollment via Beckn ONIX + TEE
- Scale to 1 utility + 1-2 aggregators; convert Fannie Mae engagement
- Begin AI-assisted edge optimization, policy-as-code



2030 Target

- 100,000 gateways deployed
- ≈ 2.5 GW under management

● FINANCIAL PROJECTIONS

What 100,000 gateways looks like.

Today — pre-revenue, 2 pilot plants, \$0 MRR, ~\$2,000/mo burn

ILLUSTRATIVE MODEL — DIRECTIONAL, NOT A FORMAL FORECAST

2030 target — 100,000 gateways, ≈2.5 GW under management

100,000 gateways

Today

2030

100,000 gateways × \$150 one-time license ≈ **\$15M** cumulative hardware/license value at full scale — illustrative arithmetic, not a revenue forecast — plus recurring SaaS layered on top. Capital-efficiency story: real pilots, ~\$2K/month burn, no fabricated growth curve.

● THE ASK

Pre-seed.
Building toward **Seed.**

Pre-seed

Current stage

~\$2K/mo

Current burn

Dec 2025

MVP live since

2 + 5

Pilot partners + DISCOM channel

TO BE FINALIZED

Raising **[\$X]** to **[reach Y milestone]** over **[Z months]**.

[HARDWARE / PRODUCTION]

[TEAM]

[PILOT SCALING]

What this round unlocks: production PCB at scale, full protocol coverage, scaled pilots (1 utility + 1–2 aggregators), and converting the active Fannie Mae engagement.

● SEE IT LIVE

**Every energy asset,
visible and
dispatchable.**

KWHELECTRIC.IO

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One gateway. Zero integration tax.